| **Surname and Initials:** | **Session Day:** | **Group no** |
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| Reyneke, R | Wed | 1 |
| **Student number:** U21495034 | | |

UNIVERSITEIT VAN PRETORIA
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# DEPARTMENT OF PHYSICS

# FSK 116

# Practical Report

## MET

## Measurements and Units

### Introduction

Please refer to the *"Lab Introductory Notes"* document for the rules you need to follow in this practical. Also read the *"Vernier Caliper Notes"* document to learn how to read the Vernier calipers. Note that your report should follow the experimental report format outlined in the Lab Introductory Notes.

### Study Aims

You must be able to:

1. Use the vernier caliper with the necessary dexterity and accuracy to make measurements.
2. Convert measurements to S.I. units.
3. Round calculated values off to the correct number of significant figures.
4. Use the experimental report format outlined on page 2 of the lab manual.

### EXPERIMENTAL AIM:

The experimental aim is to learn how to use a vernier caliper effectively and accurately to make measurements, to convert measurements to S.I. units, to round values off to the right amount of significant figures and to use the experimental format.

**RESULTS**

## A. Significant figures

Indicate the number of significant figures in the following values:

a. $9.3 \times 10^3$ km
b. 0.02600 m/s
c. 700 kg
d. What is the difference between the following two measurements: 3 km and 3.0 km?

a) 2 significant figures
b) 4 significant figures
c) 1 significant figures
d) 3 has 1 significant figure and 3.0 has 2 significant figures

Use the densities in Table 1 for the following calculations. The values in the table are also measurements so their significant figures need to be taken into account in your final answer. Work in S.I. units and round your final answers off to the correct number of significant figures.

e. Calculate the mass of a bakelite plate with dimensions 43.0 cm x 62 mm x 1.000 m

~~$\rho = \frac{m}{V}$~~
~~$1330 \cancel{\pm 33} = \frac{m}{\cancel{(43.0 \times 62 \times 1.000)}\ 0{,}027}$~~
~~$m = \cancel{0{,}0\,84}$ Kg~~
~~35,91~~

$V = 0{,}430 \times 0{,}062 \times 1.000$
$= 0{,}027$

$\rho = \frac{m}{V}$
$1330 = \frac{m}{0{,}027}$
$m = 35{,}4\cancel{6}$ Kg

f. Calculate the mass of a spherical perspex bead with a diameter of 17.0 mm.

~~$\rho = \frac{m}{V}$~~
~~$1{,}18 = \frac{m}{\cancel{V}\ 0{,}0712}$~~
~~$m = 0{,}08$ Kg~~

$V = \frac{4}{3}\pi r^3$
$= \frac{4}{3}\pi(\cancel{0{,}017})^3$ $8{,}5 \times 10^{-3}$ $= 2{,}572 \times 10^{-6}\ m^3$

$\rho = \frac{m}{V}$
$1180 = \frac{m}{2{,}572 \times 10^{-6}}$
$m = \cancel{3{,}0 \times 10^{-4}}$ $3{,}0 \times 10^{-3}$ Kg

*Table 1 The densities of various material*

| Substance | Density ($\times 10^3$ kg/m$^3$) |
|---|---|
| air | 0.00120 |
| wood | 0.3 to 0.9 |
| ice | 0.92 |
| polyethylene | 0.880 to 0.940 |
| water | 1.00 |
| blood | 1.06 |
| perspex | 1.18 |
| bakelite | 1.33 |
| PVC | 1.3 to 1.45 |
| bone | 1.7 to 2.0 |
| teflon | 2.20 |
| aluminium | 2.70 |
| glass | 2.5 to 4.2 |
| iron | 7.87 |
| stainless steel | 7.75 to 8.1 |
| brass | 8.40 to 8.73 |
| copper | 8.96 |
| silver | 10.5 |
| lead | 11.34 |
| mercury | 13.6 |
| uranium | 19.1 |

**B. Vernier caliper readings**

1. Assume that the pointer in figure 1 slid along the ruler from the 0 mark up to the indicated position. How far did it move along the ruler? Include an estimate for the fraction of the millimeter distance, Δ, in your value for d. Give your answer in millimeters.

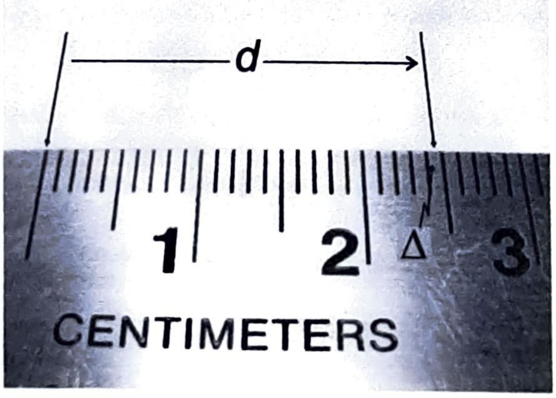
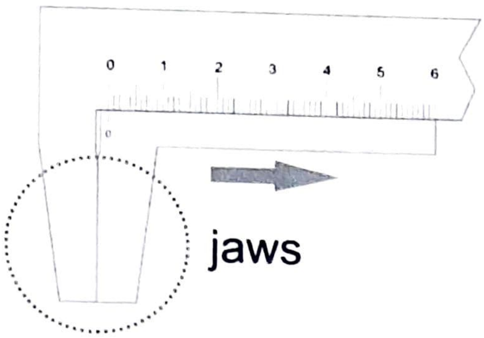



*Figure 1 The distance a pointer moved through*

| |
|---|
| 1.) 24,5 mm |
| |



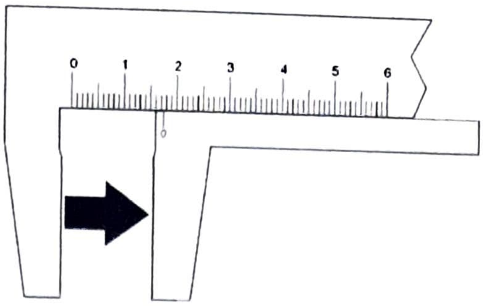




*Figure 2 Vernier caliper with jaws closed* *Figure 3 The Vernier caliper with its jaws opened*

2. The jaws of the caliper are closed in **figure 2**. The bottom jaw slides along the main cm scale just above it – see **figure 3**. The 0 mark on the bottom jaw and the 0 mark on the cm scale are aligned with each other when the jaws are closed – **figure 2**. In **figure 3** the jaws were opened to the indicated position. How many millimeters were they opened, that is how far did the 0 on the bottom jaw slide along the cm scale above it? Estimate the fraction of the millimeter value, $\Delta$, again.

2.) 17,5 mm

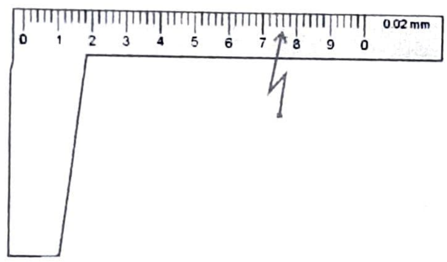



*Figure 4 The movable jaw of the caliper with the engraved vernier scale indicated on it.*

3. **Figure 4** shows the movable jaw of the vernier caliper again. On it is an engraved scale where each small division represents 0.02 mm. We say that the least count or resolution of the caliper is 0.02 mm. The line to the right of the 0 line is thus the 0.02 mm line. To its right is the 0.04 mm line, then the 0.06 mm followed by the 0.08 and the 0.10 mm with a 1 digit indicated next to it. What vernier line is the arrow pointing to on the scale in **figure 4**?

3.) 0,76 mm

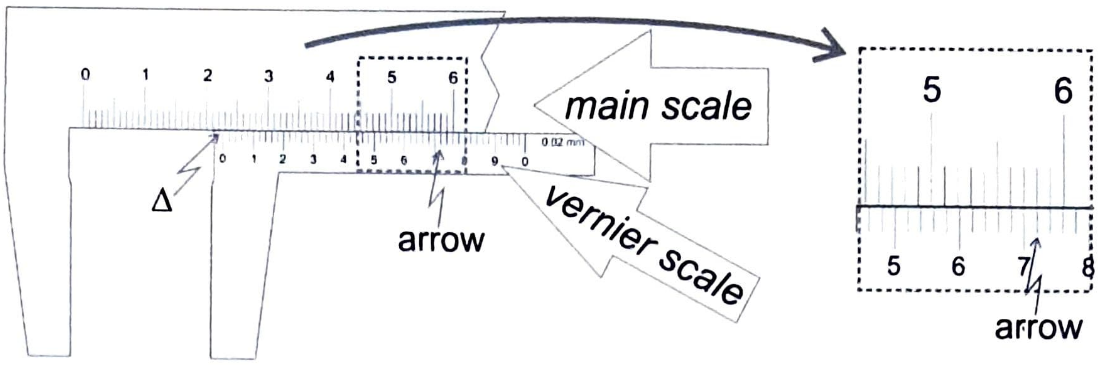



*Figure 5 Reading the vernier scale of the caliper*

4. The vernier scale enables one to accurately determine the value of, Δ, the amount by which the 0 on the vernier scale went beyond the millimeter mark on the top, main or cm scale – the Δ bit in **figure 5**. The accurate value of Δ is read off on the vernier scale and added on the reading made on the main scale. In order to determine the value of Δ we compare the position of the lines on the vernier scale to the position of the lines on the main scale - **figure 5**. Start at the left-hand end of the vernier lines and move your eye along them to the right – **figure 5**. Notice that at some point the two sets of lines come into alignment with one another - at the arrow's position. The vernier scale is read off at this alignment point, the position of the arrow. What is the vernier scale reading at this alignment point? Add this fraction of a millimeter value to your reading on the main scale - where the 0 of the vernier scale is located on the main scale - and note the final value down in your book. Note that a reading made with such a vernier caliper must always have a figure for the 1/100th of a millimeter digit! Even if the reading falls exactly on a millimeter mark, in which case you will have a 0 in the tenth and a 0 in the hundredth of a millimeter position in the reading.

| |
|---|
| 4.) 22,72 mm |
| |

5. What is the vernier caliper reading in **figure 6**?

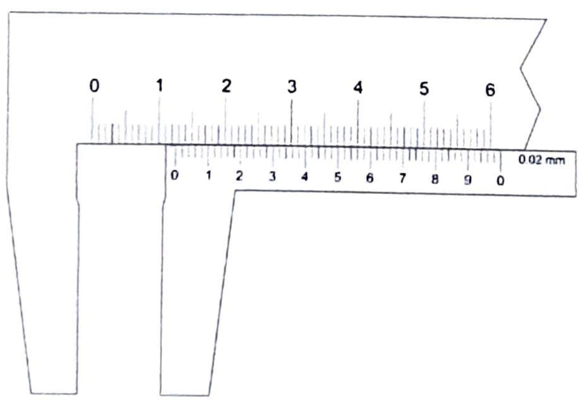



*Figure 6 Another reading on a vernier caliper*

| |
|---|
| 5.) 12,56 mm |
| |

6. The least count or resolution of the vernier caliper in **figure 7** is 0.05 mm, not like the previous examples that were 0.02mm. In this case the vernier divisions represent 0.05 mm. What is the reading in this case?


*Figure 7 A vernier caliper with a 0,05 mm resolution*

6) 14,75 mm

7. Your demonstrator will give you a simulated image of a vernier caliper. What is the reading in that case?

7.) 58,65 mm

***Below are two links to a virtual simulator for the vernier calipers. You can try the simulation yourself and you can confirm your reading.***

https://iwant2study.org/ospsg/index.php/interactive-resources/physics/01-measurements/5-vernier-caliper

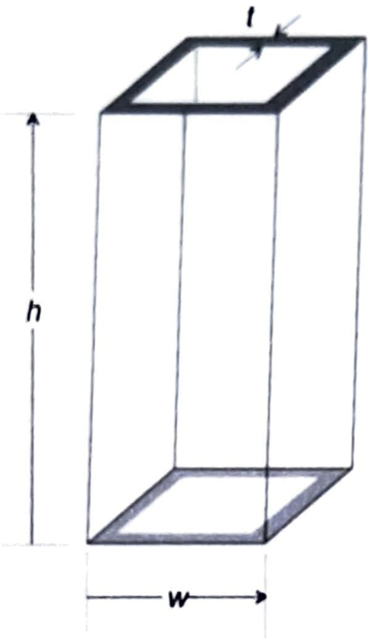
https://www.stefanelli.eng.br/en/virtual-vernier-caliper-simulator-05-millimeter/#swiffycontainer_2

## C. Density determination

To determine the density, $\rho$, of the material of a piece of metal square tubing. The density can be calculated from

$$\rho = \frac{m}{V}$$

where $m$ is the mass of the tube and $V$ its volume.




1. Set up a formula for density, $\rho$, of the hollow tube using the indicated symbols on the figure.

$$\rho = \frac{m}{V}$$

$$= \frac{m}{w^2h - h(w-2t)^2}$$

$$= \frac{m}{w^2h - h(w^2 - 4tw + 4t^2)}$$

$$= \frac{m}{h(4t^2 - 4tw)}$$

2. The dimensions of the tube were measured using Vernier calipers. Each dimension was measured 3 times, and the following measurements were obtained:

- w - 38.06mm; 38.14mm; 38.00mm
- h - 92.76mm; 92.88mm, 92.70mm
- t - 3.00mm; 3.02mm, 3.10mm

Tabulate these values and include a row for the average of each of the dimensions.

*Table heading:* Dimentions w, h, t, measured using Vernier calipers (mm)

| | | | |
|---|---|---|---|
| w | h | t | |
| 38,06 | 92,76 | 3,00 | |
| 38,14 | 92,88 | 3,02 | |
| 38,00 | 92,70 | 3,10 | |
| 38,07 | 92,78 | 3,04 | Average |

3. The mass of the tube was measured as 100.14 grams. Calculate the density of the hollow tube. Round the value off to the correct number of significant figures and write down the units of the value.

$$\rho = \frac{m}{V}$$

$$= \frac{0{,}10014}{0{,}09278\left(4\times(3{,}04\times10^{-3})^2 - 4\times3{,}04\times10^{-3}\times0{,}03807\right)}$$

$$= 2533{,}84186$$

$$\approx 2530\ kg\cdot m^{-3}$$

$$\approx 2{,}53\times10^{3}\ kg\cdot m^{-3}$$

4. Look in **Table 1** for materials with similar densities. What can you conclude from this?

The square tube is made of glass

**CONCLUSION:**

I learned how to use a vernier caliper effectively and accurately to make measurements, convert units into SI units, rounded off to the correct number of significant figures, all conveyed by using the experimental format in the lab manual.